

Md Shaqib Hussain

Project Engineer

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PROFESSIONAL SUMMARY

Innovative AI/ML Engineer with a Post Graduate Diploma in Big Data Analytics (AI & ML) from CDAC (86.88%) and a B.Tech in Computer Science Engineering (CGPA: 7.54). Experienced in designing, training, and deploying machine learning and deep learning models at scale – including CNNs, Transformer-based models, and Large Language Model (LLM) applications. Proficient in Python, TensorFlow, Keras, Scikit-learn, and PyTorch-compatible workflows, with hands-on experience in Generative AI, Retrieval Augmented Generation (RAG), prompt engineering, transfer learning, and ML model deployment via Docker, FastAPI, and AWS. GATE 2025 qualifier. Driven to build intelligent, production-ready AI systems that solve complex real-world challenges.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Object-Oriented Programming (OOP), Data Structures and Algorithms

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Ensemble Learning (Random Forest, XGBoost, LightGBM, Gradient Boosting)

Deep Learning & Neural Networks: TensorFlow, Keras, Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), LSTMs, Transfer Learning, Transformer Architecture, Attention Mechanisms, Model Fine-Tuning

Generative AI & NLP: Large Language Models (LLMs), Prompt Engineering, Retrieval Augmented Generation (RAG), LangChain, NLP Pipelines, Text Classification, Named Entity Recognition (NER), Semantic Search, Embeddings, Vector Databases

AI Model Deployment: FastAPI, Streamlit, Docker, REST APIs, Model Serving, Model Monitoring

Statistical Modeling & Analysis: Pandas, NumPy, statsmodels, Hypothesis Testing, Statistical Analysis, A/B Testing, Probability Theory, Time Series Analysis

Databases: MySQL, SQL Aggregations, Joins, Window Functions, Data Warehousing

Cloud & Infrastructure: Amazon Web Services (AWS) – EC2, S3; Linux, Shell Scripting

Data Visualization: Tableau, Power BI, Matplotlib, Seaborn

Tools & Workflow: Git, GitHub, VS Code, JupyterLab, Microsoft Excel (Advanced)

EXPERIENCE

Graduate Engineer Trainee (Data Analyst)

Mar 2026 – Aug 2026

SINCLUS Engineering & Consulting Pvt. Ltd. — BPCL Project

Mumbai, India

- Collected, cleaned, and prepared data from multiple sources to support analysis, reporting, and business requirements.
- Used SQL and Excel to extract, transform, analyze, and organize data for operational reporting and decision-making.
- Analyzed datasets to identify trends, patterns, and key insights, supporting data-driven business decisions.
- Created reports and dashboards to visualize key metrics and communicate analytical findings to stakeholders.
- Collaborated with cross-functional teams to understand data requirements and provide timely analytical insights.

PROJECTS

IntelliScan: AI-Assisted Cancer Screening System | TensorFlow, Keras, CNN, TransferLearning, Scikit-Learn Feb 2026

- Designed and trained a CNN-based deep learning model using TensorFlow and Keras for automated binary cancer detection and multi-class cancer type classification across 5+ cancer categories.
- Applied transfer learning (VGG-style architecture) and systematic hyperparameter tuning to improve model generalization on imbalanced classes, achieving 91% validation accuracy and 0.93 AUC-ROC.
- Implemented data augmentation, dropout regularization, and early stopping to prevent overfitting; evaluated outputs using precision, recall, F1-score, and confusion matrices – lifting classification accuracy by 18%.
- Leveraged Apache Spark (PySpark) for distributed preprocessing and feature extraction on 50,000+ images, enabling reproducible and scalable deep learning workflows on high-volume datasets.

Air Quality Forecasting System | Python, Scikit-learn, LightGBM, XGBoost, statsmodels, Streamlit, AWS EC2 Jan 2026

- Built an end-to-end ML pipeline for multi-hour AQI forecasting – covering data ingestion, exploratory data analysis (EDA), time-series feature engineering, model training, evaluation, and AWS deployment.
- Engineered lag variables, rolling statistics, and interaction features using Pandas and statsmodels to capture complex temporal dependencies across 5+ air pollutant types.
- Trained and cross-validated LightGBM and XGBoost ensemble models across 10+ forecast horizons, achieving R² of 0.89 and RMSE of 12.4 – a 22% improvement over the baseline.
- Deployed an interactive Streamlit application on AWS EC2 enabling real-time AQI forecasts from user-uploaded CSV data, validated by 20+ test users during evaluation.

EDUCATION

Post Graduate Diploma in Big Data Analytics (AI & ML)	Percentage: 86.88%
CDAC – Centre for Development of Advanced Computing, Pune	Aug 2025 – Feb 2026
Bachelor of Technology in Computer Science and Engineering	CGPA: 7.54
Heritage Institute of Technology, Kolkata	Aug 2021 – Jun 2025

CERTIFICATIONS & ACADEMIC HIGHLIGHTS

CDAC PG Diploma in Big Data Analytics (AI & ML) – 86.88%	Feb 2026
GATE 2025 – Computer Science and Engineering (Qualified)	Mar 2025
Microsoft Excel: Basic to Advanced	2026

EXTRACURRICULAR ACTIVITIES

National Service Scheme (NSS)	Aug 2021 – Jun 2025
Coordinator – Heritage Institute of Technology, Kolkata	
- Led campus and community awareness drives; coordinated educational support programs in local schools in Kolkata, benefiting 100+ students over 4 years.	